

MAKAR BADERKO

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ABOUT ME

I'm a 17-year-old software engineer and AI researcher with an interdisciplinary background in mathematics, computer science, economics, and physics. My work focuses on machine learning and data-driven modeling.

SKILLS

ML & AI	PyTorch, Classic ML, Gradient Boosting, Data Mining, Time-series Modelling, Deep Learning, Computer Vision, LLM Fine-Tuning, Graph Neural Networks
Quantitative Economics	Mathematical Modelling, Statistical Experiment Design, Stochastic Optimisation
Languages	Econometric Analysis, Financial Modelling
	Russian – Native, English – C1 (Cambridge certified), German – B2

RESEARCH

- AdLoCo: Adaptive batching significantly improves communications efficiency and convergence for Large Language Models - *ICOMP 2025*

ACADEMIC ACHIEVEMENTS

Computer Science and AI

- National Statistics and Data Analysis Olympiad - Gold(2022, 2024, 2025), Silver(2023)
Top-1 result in the theoretical section (2022, 2024) & all-time national cumulative score record holder
This is the most significant olympiad in data analysis in Russia
- Kaggle Competitions Master – world's top 300 Data Scientist (1x Gold, 2x Silver)
- Autumn National Olympiad in AI (IAIO 2026 National Qualifying Round) – 5th in the country
- National Technological Olympiad in Artificial Intelligence – Gold(2025)
Top-1 result (among my grade participants)
This was the most significant olympiad in AI in Russia at the time of participation

Eligible for merit-based admission to Russia's top-tier universities (MIPT, HSE, MSU, ITMO)

WORK EXPERIENCE

Laboratory of Mathematical Methods of Optimisation, MIPT

Machine Learning Research Intern

(June 2025 - August 2025)

- Co-developed AdLoCo, a novel distributed optimisation framework that significantly improves communication efficiency and convergence for LLMs
- Contributed to the mathematical proof of communication complexity bounds for non-convex stochastic optimisation

Educational Scientific Center Sirius

Machine Learning Research Intern

(July 2024 - October 2024)

- Applied Hydra bidirectional and Mamba/Mamba2 architectures to the problem of sequential recommendation, outperforming SOTA models in latency & NDCG
- Managed model fine-tuning on a GPU cluster, utilizing complex custom-developed schedulers and adaptive batching techniques

Granted direct admission to internships in ML at largest russian banks (Sber and Tbank) for winning top corporate ML competitions (interview waived).

PROJECTS

Computer Science and AI

- Machine learning model that uses publicly available blockchain network data to prevent a transaction with a high risk of money laundering (GNNs and more)
- Parcel weight and size estimation by photo and description using large language models and computer vision techniques for Avito, the largest classified ads platform in Russia
- NCAA Basketball matches outcome estimation using past match performance, bookmaker odds and machine learning (Kaggle Gold Medal)

Economics and Finance

- Governmental investment effectiveness evaluation and prediction based on overall economy performance using data about 200,000 companies
- Private investment regularity evaluation and prediction for brokerage recommender systems using data about 5000 clients